GENOMELENS

Economic analysis of a genomic report

Cost–utility model of acting on the findings in one whole-genome input, valued at $100,000 per QALY from a healthcare sector perspective.

Source input	**public benchmark genome hu4AE0F2**	Input type	**WGS**
Generated	**24 October 2026**	Time horizon	**lifetime for the carrier models**
Discounting	**3.0%, costs and QALYs**		

Proprietary.
Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.

Public benchmark genome.
A public benchmark genome. It is a real person's genome; the findings are shown to demonstrate the analysis and are not a clinical report.

Research and educational use — not medical or financial advice.
This is a decision-analytic model, not a clinical or economic evaluation submitted to any authority. Its parameters are population averages applied to one genome: it values the expected consequences of acting on a pattern of findings, and does not predict what will happen to any individual. Monetary figures include health valued at a willingness-to-pay threshold and are not cash returned to anyone. Clinically relevant findings require confirmation by an appropriate professional — a physician, clinical pharmacist or board-certified genetic counselor — before anything follows from them.

Economic impact overview

Modeled economic impact

MODELED NET MONETARY BENEFIT (SCENARIO)

$41,608

PRIMARY MODELED NET MONETARY BENEFIT UNDER THE STANDARDIZED AGE-45 SCENARIO

ASSUMPTION-DRIVEN SCENARIO

MSH2 carrier identified (man)

Primary decision-relevant comparison: colonoscopy every 2 years to 75, against no surveillance (an assumed comparator: an unidentified carrier would receive average-risk screening, so surveillance against none overstates the increment). The secondary published-strategy comparison is on the methodology page.

Model basis: hypothetical: built from published components and stated assumptions, not a published model's result.

+0.4372 QALYs	+$2,114	Assumed age 45	$100,000 / QALY
HEALTH GAIN	INCREMENTAL COST	SCENARIO	THRESHOLD

MODEL EVIDENCE
Limited: not evidence-supported even at an observed age

DONOR APPLICABILITY
Not established with available context: the donor's age is not observed

Net monetary benefit: the QALYs gained valued at $100,000 each, less the change in healthcare cost; a health-economic value, not cash saved.

perspective: US, health-care sector; horizon: lifetime, to 80; costs in 2019 US dollars; 5th to 95th percentile of the draws (parameter uncertainty only) $36,926 to $48,861.

Cost-effectiveness: $4,835 per QALY gained, below the $100,000 threshold.

Estimand: returning an already generated result. Genome-first screening value, a different question (not netted, not added): a Lynch-only test at 200 USD per person is not cost-effective in Guzauskas 2022 (ICER 153,200 USD per QALY without cascade testing), so its net monetary benefit is negative per person screened and per carrier found; the value above starts after sequencing, so the test cost is sunk.

Other modeled values (shown separately, never added)

ALTERNATIVE MODELED SCENARIO

ASSUMPTION-DRIVEN SCENARIO

$40,160

Lynch syndrome carrier identified → colonoscopy surveillance after identification. Not included in the primary value; it fits this person less well (age at identification)

FAMILY / CASCADE-TESTING VALUE

DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO

$7,205

Lynch syndrome carrier identified → colonoscopy surveillance after identification. Different beneficiary (relatives); not additive to the individual value

SUPPORTED CALCULATION AT THE ASSUMED AGE

None supported — No decision has every model input sourced at the assumed age, so no value is supported there (an empty set, not a measured zero). This strict section admits only decisions whose every input is sourced at the standardized age of 45; it does not replace the modeled values shown with their own classification.

Research and educational use only; not medical, clinical or financial advice. Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Its assumptions are listed on the economic methodology sheet.

All economically relevant findings

Every economic result of this genome, ranked within its class. Values in different classes are never added.

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
MSH2 carrier identified (man)	colonoscopy every 2 years to 75	**$41,608**	PRIMARY MODELED ASSUMPTION-DRIVEN SCENARIO	Counted: the selected model's expected value
Lynch syndrome carrier identified	colonoscopy surveillance after identification	**$40,160**	ALTERNATIVE MODELED ASSUMPTION-DRIVEN SCENARIO	Not counted: it fits this person less well (age at identification)
Lynch syndrome carrier identified (Bennette 2015)	return of the finding, then guideline management	**$90,492**	ALTERNATIVE MODELED ASSUMPTION-DRIVEN SCENARIO	Not counted: its penetrance comes from family-based estimates not corrected for ascertainment, so it likely overstates the value to a carrier found by a genome-first screen
Lynch syndrome carrier identified	colonoscopy surveillance after identification	**$7,205**	FAMILY / CASCADE-TESTING VALUE DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO	Accrues to relatives; a different beneficiary, not added
Hypothetical, based on assumptions: any adult, before results	Population screening for Lynch syndrome, hereditary breast and ovarian cancer and familial hypercholesterolemia	**per patient tested**	TESTING-STRATEGY / POPULATION-LEVEL VALUE	A population testing-strategy average per patient tested, not this genotype's value
ANKK1		**valued at zero**	TRUE ZERO	ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for …
DRD2		**valued at zero**	TRUE ZERO	ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for …
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)		**valued at zero**	TRUE ZERO	Not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified …
vitamin D (at age 45)		**valued at zero**	TRUE ZERO	Outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older …

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

All economically relevant findings (continued)

FINDING	DECISION PATHWAY	MODELED NMB	EVIDENCE CLASSIFICATION	KEY LIMITATION
CBS pathogenic (ClinVar)		**not quantified**	UNQUANTIFIED	No economic model yet
NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing		**not assessable**	TECHNICALLY UNRESOLVED	NAT2 is inferred here from three markers (*5/*6/*7) without phase

The evidence hierarchy

- **Evidence-supported**: every model input is sourced and the value does not rest on an age that is stated rather than observed (badge EVIDENCE-QUALIFIED)
- **Primary modeled**: the selected economic model for a decision; it may depend on the stated assumptions shown beside it **in this genome**
- **Once the decision arises**: the value per person if the decision arises; not an expected value today
- **Alternative modeled**: a competing model or sensitivity scenario for the same decision; never added to the primary result **in this genome**
- **Conditional**: valid only if a stated clinical or treatment condition holds; never added
- **Family / cascade-testing**: accrues to relatives or another beneficiary; never in the individual's personal total **in this genome**
- **Population / testing strategy**: a value per person screened or tested in a population, not this donor's value; never added **in this genome**
- **Hypothetical / projected**: an exploratory scenario on stated assumptions, not an observed or established benefit; in no total **in this genome**
- **Negative**: modeled, and the costs exceed the health gain valued at $100,000 per QALY; shown, never hidden
- **True zero**: the evidence supports no incremental benefit or change in care within the defined decision **in this genome**
- **Unquantified**: potentially relevant, but the evidence is insufficient for a monetary value; not $0 **in this genome**
- **Technically unresolved**: the genomic evidence cannot reliably support the interpretation the decision needs; not $0 **in this genome**
- **Not applicable**: the model or intervention does not apply to the assessed context (sex, age or setting); not $0

Unknown is not $0. Unquantified is not $0. Technically unresolved is not $0. No supported estimate is not a measured zero. Negative is not zero. An alternative is never added.

Economic methodology: how the primary modeled value is calculated

FINDING
MSH2 carrier identified (man)

PRIMARY DECISION-RELEVANT COMPARISON
colonoscopy every 2 years to 75; compared with no surveillance

MODEL EVIDENCE
Limited: not evidence-supported even at an observed age

DONOR APPLICABILITY
Not established with available context: the donor's age is not observed; the figure is the standardized age-45 scenario, a standardized calculation, not a clinical prediction for this donor

CHANGE IN HEALTH OUTCOME
+0.4372 QALYs; +1.152 life-years, undiscounted

CHANGE IN COST
+$2,114 (costs in 2019 US dollars; positive is more spending)

ECONOMIC CALCULATION
0.43722 QALYs times $100,000 per QALY, minus the change in cost of $2,114

RESULT

$41,608 ASSUMPTION-DRIVEN SCENARIO

Primary modeled net monetary benefit

Source and scope

population: a man with an incidental MSH2 finding; valued per: carrier identified; setting: US, health-care sector perspective (Medicare-based costs); horizon: lifetime, to 80; sources: PMID 35809086; PMID 33839100. Net monetary benefit is the QALYs gained valued at $100,000 each, less the change in healthcare cost, discounted at 3% a year unless the row names its source's own rate.

Key assumptions

- at the stated age of 45, assumed for this run, not observed
- no surveillance: an assumption of this model, not the source's comparator arm
- comparator: no surveillance; age stated, not observed
- the 2-yearly effect (hazard ratio 0.535) is the source's assumption: 0.05 below its fitted 3-yearly 0.585

Secondary published-strategy comparison

Kastrinos 2021, PMID 33839100, Table 2 compares colonoscopy schedules among carriers who are already known, identified at 25, with perfect adherence, in 2019 US dollars. It has no 'not identified' arm, so it is not this decision's value and is in no total. The source's optimal schedule for MSH2 (every 2 years from age 25) against the next efficient schedule (every 3 years from age 25): +0.028 QALYs, +908 USD, a net monetary benefit of 1,892 USD at 100,000 USD per QALY. Against the least intensive schedule it lists (every 5 years from age 40, current management for a family history, which the source rates dominated): +0.664 QALYs, −25,131 USD, a net monetary benefit of 91,531 USD. One schedule against two comparators: the figures are not interchangeable. Its discounting is unresolved: the printed QALYs exceed what its stated 3% can give, so these figures are arithmetic on the totals as printed.

Where multiple economic models are available, GenomeLens evaluates evidence relevance, population applicability, intervention comparability, parameter provenance, and uncertainty. Alternative supported models may be retained as sensitivity scenarios.

Findings valued by their decision models

Net monetary benefit (NMB) is the QALYs gained valued at $100,000 each, less the change in healthcare cost, in 2019 US dollars. It is a health-economic value, not cash saved: a positive NMB can come with higher spending. Each clinical decision is valued once; the categories are never added to each other. Each figure's horizon is its own model's (the carrier models run over a lifetime; no pooled disease model ran for this genome).

Standardized economic scenarios: age 45, using genomic findings identified from publicly available research datasets. Genomic calls in this proof of concept have not been independently confirmed by a clinical laboratory and are not diagnostic results. Age 45 is stated for this run, not observed (one demonstration age for every sample; the PGP public profiles give only an age bracket, and the GIAB and 1000 Genomes samples publish no age). A value whose model depends on age is modeled at that age; a value its source gives for one age only is held at that age, and its row says so. Model evidence is stated apart from whether its context matches the donor, and the qualification stays assumption-driven; an age-dependent value would be evidence-qualified only at an observed or self-reported age.

Sex: known male.

Supported calculation at assumed age 45

None supported AGE 45 ASSUMED No decision has every model input sourced at the assumed age, so no value is supported there (an empty set, not a measured zero). The values limited by inputs besides the age are in B, never added here. This is the stricter test: the modeled values, which rest on stated assumptions, are shown with their own class and are not contradicted by it.

- no decision is supported at the assumed age

A. Donor-specific economic value

Not established with available context — the donor's age is not observed, and 1 priced decision(s) depend on it; no total is stated for this donor, which is not $0.

- none

B. Other modeled expected value, with further limitations

Further-limited subtotal $41,608 ASSUMPTION-DRIVEN SCENARIO — limited by inputs besides the age; never added to the supported calculation above.

- **MSH2 carrier identified (man): colonoscopy every 2 years to 75 (GenomeLens model, hypothetical)**: $41,608
  ASSUMPTION-DRIVEN SCENARIO

  ΔQALY +0.4372; ΔCost $2,114 (2019 USD; positive is more spending); ΔLY +1.152, undiscounted; 5th to 95th percentile of the draws $36,926 to $48,861 (parameter uncertainty only).

  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.

  Holds at the stated age of 45, assumed for this run, not observed; if identified at 45.

  Limited by: no surveillance: an assumption of this model, not the source's comparator arm; comparator: no surveillance; age stated, not observed.

  comparator: no surveillance; horizon: lifetime, to 80; sources: PMID 35809086; PMID 33839100.

C. Values that hold only in a stated clinical context

- none for this genome

Findings valued by their decision models (continued)

D. Alternatives and scenarios, shown separately

- *Alternative modeled scenario, not counted: it fits this person less well (age at identification)*: **Lynch syndrome carrier identified: colonoscopy surveillance**: $40,160 ASSUMPTION-DRIVEN SCENARIO
  ΔQALY +0.5091; ΔCost $10,754 (2019 USD; positive is more spending); ΔLY not reported by the source.
  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.
  Holds at the stated age of 45, assumed for this run, not observed; the source's value for identification at 30, held at 45.
  Limited by: age at identification; age stated, not observed.
  comparator: family history-based testing alone (the status quo); horizon: lifetime; sources: PMID 35227606.
- *Alternative modeled scenario, not counted: its penetrance comes from family-based estimates not corrected for ascertainment, so it likely overstates the value to a carrier found by a genome-first screen*: **Lynch syndrome carrier identified (Bennette 2015)**: $90,492
  ASSUMPTION-DRIVEN SCENARIO
  ΔQALY +0.95; ΔCost $4,508 (2019 USD; positive is more spending); ΔLY not reported by the source.
  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.
  Holds at the stated age of 45, assumed for this run, not observed; if identified at 45.
  Limited by: penetrance not corrected for ascertainment; its source model was later corrected downward; age stated, not observed.
  comparator: the finding not returned; horizon: lifetime; sources: PMID 25394171.
- *accrues to relatives, not to the person; in no personal total*: **Lynch syndrome carrier identified: colonoscopy surveillance: cascade testing of relatives**: $7,205 DIFFERENT BENEFICIARY · ASSUMPTION-DRIVEN SCENARIO
  ΔQALY +0.11; ΔCost $3,795 (2019 USD; positive is more spending); ΔLY not reported by the source.
  Model evidence: limited: not evidence-supported even at an observed age; **donor context:** not established with available context: the donor's age is not observed.
  Holds at the stated age of 45, assumed for this run, not observed.
  Limited by: age at identification; age stated, not observed.
  comparator: family history-based testing alone (the status quo); horizon: lifetime; sources: PMID 35227606.

E. Actionable pathways not yet quantified

- CBS pathogenic (ClinVar) — actionable economically unquantified: no economic model yet.
- NAT2 slow acetylator (three-marker inference): NAT2-guided isoniazid dosing — technically unresolved: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a prescribing-grade genotype, so no drug decision is valued. Missing input: a prescribing-grade NAT2 genotype.

F. Valued at zero: care does not change (a result, not a gap)

- **ANKK1** — valued at zero: ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for (https://ispg.net/genetic-testing-statement/)
- **DRD2** — valued at zero: ISPG 2019: common psychiatric risk variants are individually of no clinical significance and should not be tested for (https://ispg.net/genetic-testing-statement/)

Findings valued by their decision models (continued)

F. Valued at zero: care does not change (a result, not a gap) (continued)

- **ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)** — valued at zero: not a pathogenic variant. ATP7B c.2495A>G p.Lys832Arg (rs1061472), the only variant behind this row, is classified Benign in ClinVar (Variation 35707) and is a common polymorphism, not a Wilson disease allele (ClinVar Variation 35707).
- **vitamin D (at age 45)** — valued at zero: outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older (LeBoff 2022, PMID 35939577); Endocrine Society 2024 (Demay, PMID 38828931): for adults younger than 50 it suggests against empiric vitamin D beyond the dietary reference intake and against routine 25(OH)D testing

Findings-first economics

No pooled result: no finding enters the pooled model, so there is no combined estimate to state; each finding is valued by its own decision model, by category, on the next page.

This report starts with what was found in the genome. The economics explains the consequence of acting on each finding; the pooled reference case shows what remains after overlapping signals, imperfect adherence and the real cost of acting are all accounted for.

HEALTH GAIN

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

HEALTHCARE COST CHANGE

Not available

no pooled result

MODELED NET BENEFIT

Not available

no pooled result

DECISION UNCERTAINTY

Not available

no pooled result

0

findings carry a standardized economic estimate

12 of the 12 findings shown carry no standardized estimate, and none is assigned one the model cannot support. Separately, 1 clinical decision(s) are valued by their own decision models (0 evidence-qualified, 1 modeled with stated assumptions), by category in the section 'Findings valued by their decision models'; a finding with such a value has its figure there, not here.

DRD2/ANKK1 DRD2 / ANKK1 Taq1A · Vitamin D Synthesis Efficiency · MSH2 Lynch syndrome variant · Hereditary Hemochromatosis · Wilson Disease · NAT2 — NAT2 slow acetylator… · Reproductive genetics · BTK pathogenic · MT-ND5 pathogenic · ETFA pathogenic · WFS1 pathogenic · CFD pathogenic

Medication-genotype findings

Where a genotype could change a prescribing decision

These are the clearest cases of a genomic result bearing on a future medication choice. The clinical decision leads; the economics explains why preserving the genotype for that decision has value.

This run produced no medication-genotype finding with a priced economic pathway; a finding whose decision the evidence values at zero is listed with the true zeros.

Decision support, not a directive. Nothing here is an instruction to start, stop or change a medication. Relevance depends on whether the drug is actually prescribed, and the prescribing decision belongs with a qualified clinician who can see the whole clinical picture.

Risk and prevention findings, in detail

No risk or prevention finding of this genome has a standardized estimate, so there is no card to show. The findings are reported without one below, each with its reason on the glance pages; a lower evidence grade changes how a finding should be read, not whether it is shown.

Reported without a standardized value (11): DRD2/ANKK1 DRD2 / ANKK1 Taq1A..., Vitamin D Synthesis Efficiency..., Hereditary Hemochromatosis (mild..., ATP7B rs1061472 (benign polymorphism)..., NAT2 — NAT2 slow acetylator..., Reproductive genetics — 2 carrier..., and 5 more. Each carries its reason on the glance pages (no change in care, not assessable, unquantified, or reproductive information); no value is invented.

How the findings combine

Taken together, what do these findings mean?

There is no pooled reference case for this genome: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. Findings bearing on one condition are combined here before any value is credited; with nothing in the pool there is nothing to combine, and no figure stands in for the absent result. The findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'.

INCREMENTAL HEALTHCARE COST

Not available

no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost; the findings are valued by their own decision models instead, by category, in the section 'Findings valued by their decision models'

INCREMENTAL QALYS

Not available

no pooled result

REFERENCE-CASE NMB

Not available

no pooled result

CLASSIFICATION

Not available

no pooled result

How sure is this, and what rests on judgment?

Uncertainty has two layers: the spread of modeled outcomes, and the provenance of the inputs producing that spread. A robust decision does not make its inputs precise, so both are shown.

Net monetary benefit across simulations

No probabilistic analysis of the pooled model: no finding enters the pooled model: every candidate row is refused or unpriced, so there is no established benefit to set against the test cost. No probability of cost-effectiveness, interval or one-way swing is stated for it, because there is no pooled strategy to sample. The decision models carry their own parameter uncertainty (2,000 draws from the ranges the sources print, rescaled to a per-carrier value where a source prints another unit); where a decision has a range, its 5th to 95th percentile is printed beside that decision's value.

Hypothetical, based on assumptions

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Base case: the value the published evidence supports for this genome, or “not estimable” with the reason. Scenario: what GenomeLens computes on its own assumptions, graded by the quality of its inputs, with each assumption numbered in the log below. Scenario figures are not from published data for this genome and are not the evidence headline; the reproductive costs belong to a prospective child and are in no personal total. A row valued by its own decision model repeats that model's value as a pointer; the model's basis (published, or hypothetical) is stated beside the headline. This genome's assessment is incomplete (a finding is actionable but unquantified, technically unresolved, or lacks the evidence to value): that is not a conclusion that the genome has no further economic value.

Table 1. Results by finding

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
This genome, to the person sequenced (each finding once)	**Not established** Incomplete assessment: not zero	**$41,608** Sum of the rows not marked 'not summed': model and scenario values, never this donor's total	**no evidence-supported value; $41,608 from decision models that are not evidence-qualified; no projected finding is summed (a placeholder on its own row is not added); figures marked 'not summed', reproductive values and payer values are excluded**
DRD2/ANKK1 DRD2 / ANKK1 Taq1A — DRD2/ANKK1 Taq1A — reward-pathway variant	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	The genotype does not change care (a guideline: care does not change)
Vitamin D Synthesis Efficiency — Nutrient-metabolism variant — deficiency prevention	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	Care does not change under a guideline (outside the trial's population at this age: VITAL enrolled men 50 and older and women 55 and older ... Sources reviewed: PMID 17634462; PMID 32533132; PMID 35727271
MSH2 Lynch syndrome variant (heterozygous)	**See its decision row** Decision model	**$41,608** Decision model (not summed)	Valued by the decision model 'MSH2 carrier identified (man): colonoscopy every 2 years to 75 (GenomeLens model, hypothetical)' (modeled with stated assumptions)
Hereditary Hemochromatosis (mild variant) carrier (modifier allele)	**Not estimable** Not estimable	**No model yet** No model yet	No earlier figure; the decision models' verdict: no value of its own: it modifies another finding: modifies another finding rather than carrying its own
ATP7B rs1061472 (a benign polymorphism, not a Wilson disease finding)	**Zero** Published evidence	**Not projected** The evidence supports no incremental benefit, so none is projected	No earlier figure; the decision models' verdict: valued at zero: not a pathogenic variant
NAT2 — NAT2 slow acetylator — occupational cancer risk awareness	**Not estimable** Not estimable	**Not estimable** Conditional on a genotype call this input cannot establish; no figure is shown	No earlier figure; the decision models' verdict: not assessable from this input: NAT2 is inferred here from three markers (*5/*6/*7) without phase, which is not a ... Sources reviewed, none of which carries a figure: PMID:16112301
Reproductive genetics — 2 carrier condition(s) where partner testing would be informative	**Separate arm** Separate arm	**See reproductive table** Reproductive table	Costed per risk in the reproductive rows (a prospective child's value, kept out of the personal total)
BTK pathogenic (ClinVar, xlinked)	**Not estimable** Not estimable	**No model yet** No model yet	Actionable; economic value not established

Hypothetical, based on assumptions (continued)

HYPOTHETICAL: base case and GenomeLens assumption scenario, by finding (hypothetical, based on assumptions)

Table 1. Results by finding (continued)

FINDING	BASE CASE (PUBLISHED EVIDENCE)	SCENARIO (GENOMELENS ASSUMPTIONS)	WHY THE BASE CASE IS WHAT IT IS
MT-ND5 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
ETFA pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
WFS1 pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section
CFD pathogenic (ClinVar, uncertain inheritance)	**Not estimable** Not estimable	**No model yet** No model yet	A pathogenic / likely-pathogenic ClinVar variant in a gene with no modeled condition anchor. It is reported in the clinical-variants section

Table 2. Reproductive risks identified, with their cost (per pregnancy, for a prospective child; ancestry used: European)

REPRODUCTIVE RISK	CHANCE	COST OF THE CHILD'S CONDITION	EXPECTED COST	TEST THAT RESOLVES IT	SOURCE OF THE COST
HFE H63D Hereditary Hemochromatosis (mild variant); recessive; carrier	**1 in 86** of an affected child, per pregnancy	**Not available** no cost of the child's condition is registered for HFE; only the test that resolves the risk is priced	**—** per pregnancy	**$65** the partner's carrier test	GenomeLens registered estimate
ATP7B K832R Wilson Disease (variable); recessive; common polymorphism, not a disease allele	**not stated** of an affected child, per pregnancy	**Not available** this allele is commonly reported as a polymorphism rather than a disease allele (classified as uncertain significance in the carrier catalog): it cannot complete an affected genotype, so neither a chance of an affected child nor a cost of the child's condition is stated; there is no risk for a partner's test to resolve, so none is priced	**—** per pregnancy	**—** No test priced: no risk to resolve	GenomeLens registered estimate

State: the figures on this page are model outputs, labeled UNRESOLVED (not established for this donor) unless a row carries another label. Standardized scenario at assumed age 45: the donor-specific economic value is not established with available context.

Methods

How these figures were produced, and how to read them

A decision-analytic model values the expected consequences of acting on each finding in this genome. The figures are model outputs under stated assumptions, kept apart from clinical outcomes.

ANALYTIC FRAME

Perspective	**healthcare sector**
Comparator	**each decision model states its own**
Outcomes	**cost, QALYs, NMB**
Time horizon	**lifetime for the carrier models**
Discounting	**3%, costs and QALYs**
Willingness to pay	**$100,000/QALY**
Currency	**USD, United States**
Uncertainty	**decision models: 2,000-draw parameter uncertainty; pooled model: no PSA (no pooled model)**

HOW TO READ EACH FIGURE

- **Evidence-supported value:** every model input is sourced and the value does not rest on an age that is stated rather than observed. Where the age is stated, as for every public genome, an age-dependent value is a standardized scenario and is labeled so.
- **Decision-model rows:** computed from published inputs with at least one assumption (for a public genome, the stated age); shown, never presented as evidence-qualified.
- **Hypothetical, based on assumptions:** what a finding could be worth if its curated assumptions held; never added to any counted total.
- **Projected, based on assumptions:** the GenomeLens scenario beside a base case that cannot be computed soundly; not the evidence headline.
- **Reproductive costs:** per pregnancy, for a prospective child, never in a personal total.

Declared limitations

- Population-average parameters are applied to one genome. The model values expected consequences for a person with this pattern of findings, not a prediction for any individual.
- Estimates are illustrative and decision-analytic. This is not a formal economic evaluation or a submission to any authority.
- Where a condition-specific baseline risk is unavailable a generic registered assumption is used; those assumptions dominate the sensitivity analysis and are labeled as such.
- Costs are restated to 2019 US dollars from each source's price year.
- Findings without a modeled decision pathway are excluded from net monetary benefit rather than assigned an invented value.
- Clinically relevant findings require confirmation by an appropriate professional before anything follows from them.

Proprietary method. Demonstration report for a public research genome. GenomeLens, its economic models and its methods are proprietary. This report shows results, not the method: it is not open-source software, not a description sufficient to reproduce the analysis, and not a license to use either.